

BASAVA HARSHA

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SUMMARY

Motivated, Enthusiastic and dedicated Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning. Proficient in Python and Java with a solid understanding of programming fundamentals and problem-solving techniques. Demonstrates strong analytical thinking, adaptability, and a commitment to continuous learning. Eager to apply technical knowledge and AI concepts to develop innovative solutions for real-world challenges.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

Kalasalingam Academy of Research & Education, Tamil Nadu (Grade Point: 8.5/10)

Expected 2027

Intermediate Education

Sri Chaitanya Mahila Jr Kalasala, Andhra Pradesh (Grade Point: 10/10)

2023

Board of Secondary Education

Sri Chaithanya High School, Andhra Pradesh (Grade Point: 10/10)

2021

SKILLS

- Programming Languages : Python, Java
- Data Analysis & Visualization Tools : Microsoft Tools, Power Bi
- Artificial intelligence : Prompt Engineering, Responsible Ai, Vibe Coding
- Software Tools : VM VirtualBox, Docker
- Soft Skills : Problem solving, time management, adaptability

PROJECTS

1. Custom AI Agent with Memory | Python, FastAPI, LangGraph, LangChain, Streamlit, Groq API

Developed a full-stack AI agent with persistent long-term memory supporting both voice and text interactions, enabling personalized conversations by remembering user preferences, facts and goals across multiple sessions. Engineered a LangGraph ReAct agent integrated with LangChain and Groq's Llama 3.3 70B model, implementing intelligent memory retrieval, automated memory storage and contextual response generation using scoped user memories. Built a FastAPI backend exposing REST APIs for chat, memory management and Whisper-based speech transcription, while developing a Streamlit frontend featuring voice input, real-time chat and an interactive memory dashboard. Designed a scalable multi-user architecture with isolated memory stores, Dockerized deployment and cloud hosting on Render and Streamlit Community Cloud, delivering a production-ready AI assistant supporting persistent conversations and 1000+ daily requests on free infrastructure.

2. AI Code Review Bot – GitHub PR Auto-Reviewer | Python, FastAPI, GitHub Webhooks, Groq API

Developed an AI-powered GitHub Pull Request review automation system that performs intelligent code analysis and posts detailed review comments directly on GitHub using Groq's Llama 3.3 70B model. Engineered a FastAPI backend integrated with GitHub Webhooks and GitHub REST APIs to automatically retrieve pull request diffs, analyze code quality and generate actionable feedback for every submitted PR. Implemented automated detection of logic errors, security vulnerabilities, performance bottlenecks, coding standard violations and best practice recommendations while providing overall approval or rejection verdicts with file-level review comments. Deployed the backend on Render with zero-cost cloud infrastructure, enabling scalable, production-ready continuous AI-assisted code review without relying on paid AI services.

3. RAG Chatbot – Chat with your PDFs | Python, FastAPI, Next.js, TF-IDF, Groq API

Developed a full-stack Retrieval-Augmented Generation (RAG) chatbot enabling users to upload PDF documents and interact with them through natural language conversations while generating responses strictly grounded in document content. Engineered a lightweight retrieval pipeline using TF-IDF indexing and PyPDF for document parsing, efficiently retrieving relevant text chunks which were processed by Groq's Llama 3.3 70B model to generate context-aware answers with exact source page references. Implemented real-time response streaming using Server-Sent Events (SSE), low-memory document indexing, document chunking and source attribution to improve transparency, accuracy and user trust. Built the frontend using Next.js and deployed the application on Vercel with a Dockerized FastAPI backend hosted on Render, delivering a scalable, production-ready AI document assistant without external vector databases or paid APIs.

CERTIFICATIONS

Introduction to Generative AI – Google Cloud

AI Fundamentals – IBM

Introduction to AI in Azure – Microsoft

Machine Learning Fundamentals – AWS